

COURSE: CS 354 — Computer Vision

Credits: 3 | Department: Computer Science

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PREREQUISITES: CS 351

CO-REQUISITES: None

OFFERED: Spring

DESCRIPTION:

Image processing and computer vision. Topics include image formation, filtering, edge detection, feature extraction (SIFT, HOG), object detection (YOLO, R-CNN), image segmentation (semantic, instance), 3D reconstruction, optical flow, and video analysis. Students implement vision pipelines using OpenCV and PyTorch.

GRADING: Lab assignments 40%, Course project 40%, Midterm 20%.